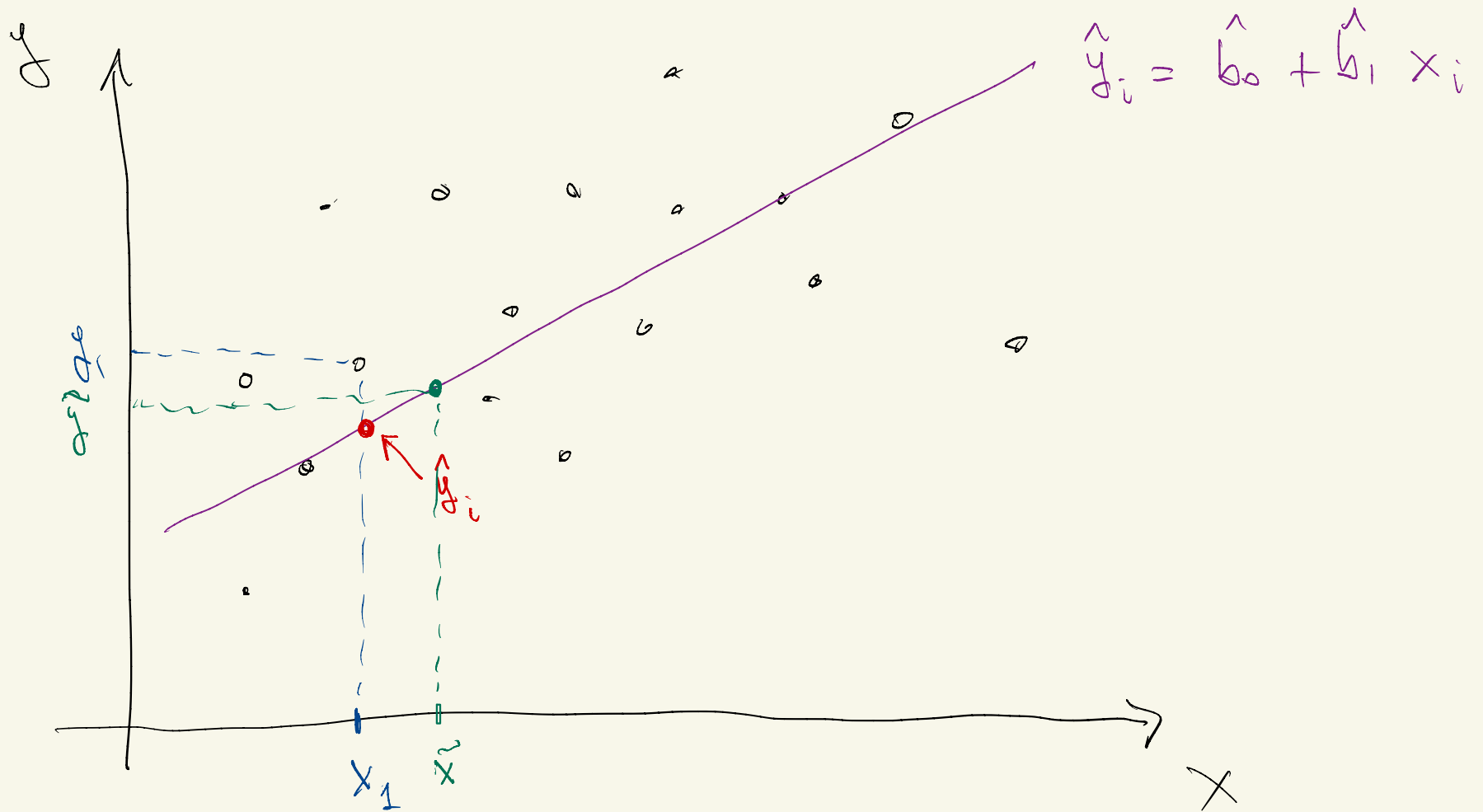


Eco 202 - Fall 2026

Class 04



$X_1 = 3$ yr of education

$y_i = 13$ dollars per hour

Recall

$y_1, \dots, y_n$

$$\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i = \underset{a}{\operatorname{argmin}} \sum_{i=1}^n (y_i - a)^2$$

least-squares

$$(\hat{b}_0, \hat{b}_1) = \underset{b_0, b_1}{\operatorname{argmin}} \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2$$

FOC

$$\frac{\partial}{\partial b_0} \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2 = 2 \sum_{i=1}^n (y_i - b_0 - b_1 x_i) \cdot (-1)$$
$$\frac{\partial}{\partial b_1} \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2 = 2 \sum_{i=1}^n (y_i - b_0 - b_1 x_i) (-x_i)$$

$$\frac{\partial}{\partial b_0} \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2 = 2 \sum_{i=1}^n (y_i - b_0 - b_1 x_i) \cdot (-1) = 0$$

$$\frac{\partial}{\partial b_1} \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2 = 2 \sum_{i=1}^n (y_i - b_0 - b_1 x_i) (-x_i) = 0$$

$\therefore$ algebra

$$\hat{b}_0 = \bar{y} - \hat{b}_1 \bar{x}$$

$$\hat{b}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$